

FINAL DELIVERY

DATA ENGINEERING · NLP · NYC

EMPIRE'S TASTE.

New York gastronomic intelligence — real-time sentiment analysis of the NYC culinary scene combining press, real reviews, and recipes.

MARIA DEL PILAR PRADILLA CELY - 20231020119

JAIRO ARTURO MOSQUERA - 20222020142

DAVID SANTIAGO ALDANA GONZALEZ - 20222020158

THE PROBLEM

NYC food signals — articles, reviews, recipes — scattered & unprocessed.

Which dishes are trending this week in NYC?

Are real diners satisfied with the restaurants the press praises?

Which recipes align with what media is covering?

How does sentiment evolve week after week?

THE SOLUTION

Automated pipeline: collect → clean → analyze → deliver. Daily. No manual intervention.

3

DATA SOURCES

10+

KPIS CALCULATED

~10 MIN

FULL PIPELINE

USER STORIES PRIORITIZED BY DASHBOARD IMPACT

PRIORITY	NEED
HIGH	TOP-RATED DISHES BY EVENT TYPE → COMPETITIVE MENU PROPOSALS
HIGH	FILTER DISH SUGGESTIONS BY DIETARY RESTRICTION (VEGAN, GLUTEN-FREE, HALAL)
HIGH	SENTIMENT TRENDS BY DISH CATEGORY — DETECT CUISINES GAINING OR LOSING POPULARITY
MED	DATA QUALITY METRICS: MISSING FIELDS, INGESTION RATE, RECORD COUNTS
MED	DISHES MENTIONED ALONGSIDE NYC NEIGHBORHOODS → LOCATION-SPECIFIC MENUS
LOW	AUTOMATED ALERTS ON SUDDEN SENTIMENT DROPS → PROACTIVELY UPDATE EVENT MENUS
LOW	COMPARE RECIPE METADATA VS. REVIEW SENTIMENT → VALIDATE RECOMMENDATION MODEL

WHERE THE DATA COMES FROM

WEB SCRAPING

EATER NY

RSS/Atom feed from NYC's most important food media outlet.

- ✓ Up to 20 articles/day
- ✓ Title, summary, author, categories
- ✓ Canonical URL for traceability
- ✓ Publication date

WEB SCRAPING + GRAPHQL

OPENTABLE

Real diner reviews via OpenTable's internal GraphQL endpoint.

- ✓ Playwright (headless Firefox)
- ✓ Automatic session — no manual cookies
- ✓ Ratings: food, service, ambiance
- ✓ Free-text diner comments

REST API

SPOONACULAR

Recipe API queried dynamically with NLP-detected trending keywords.

- ✓ Free-text or ingredient search
- ✓ Nutritional & dietary metadata
- ✓ Health score, price per serving
- ✓ 6 API keys with automatic rotation

EATER NY

20 ART/DAY · RSS

+

OPENTABLE

~200 REVIEWS · GRAPHQL



NLP → KEYWORDS

TF-IDF + SPACY VECTORS



SPOONACULAR

5 RECIPES/KEYWORD

MEDALLION ARCHITECTURE

BRONZE

RAW DATA — EXACTLY AS RECEIVED

Raw JSON · date-partitioned · dedup by ID · full fidelity



SILVER

CLEAN DATA — READY FOR ANALYSIS

Parquet · HTML stripped · NLP texts · types cast
· cross-source dedup



GOLD

ANALYTICAL DATA — READY FOR CONSUMPTION

Parquet (Spark coalesce) · VADER sentiment ·
TF-IDF keywords · governance KPIs · 10+
storytelling aggregations

WHY 3 LAYERS?

Each layer has a purpose: Bronze = never lose raw data. Silver = make it usable. Gold = dashboard-ready. Recomputable from Silver if needed.

BRONZE FORMAT

JSON

SILVER/GOLD FORMAT

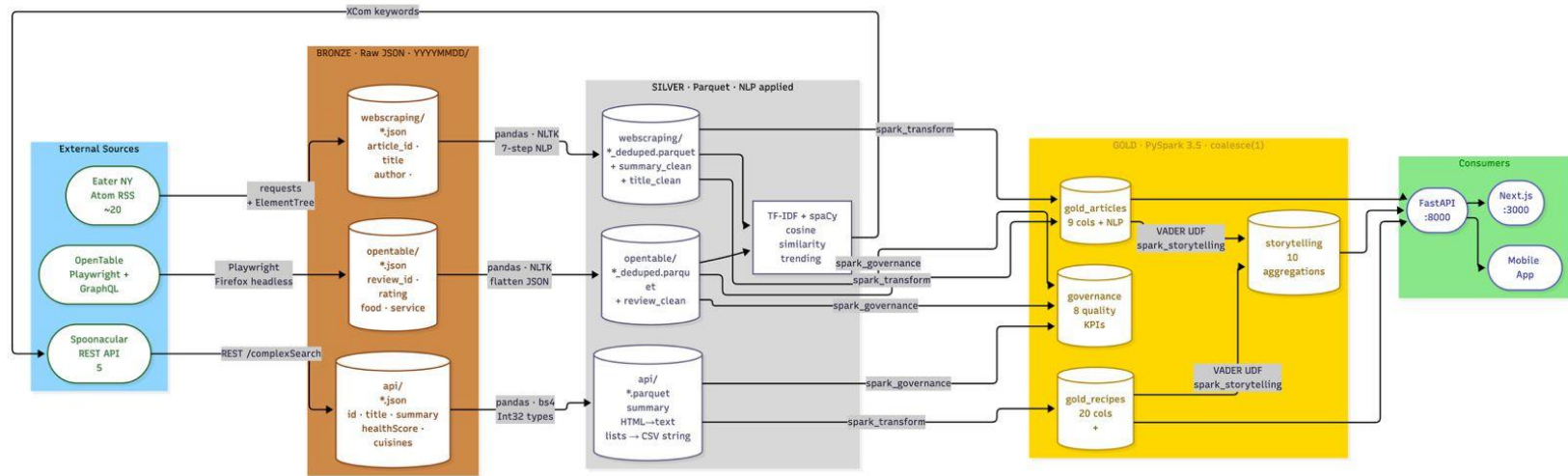
PARQUET

GOLD ENGINE

PYSPARK 3.5

ORCHESTRATION

AIRFLOW 2.10



THE PIPELINE — STEP BY STEP

DAG: NYC_GASTRONOMY_PIPELINE · MANUAL TRIGGER · SEQUENTIAL · ~10 MIN END-TO-END

#	AIRFLOW TASK	WHAT IT DOES	INPUT	OUTPUT
01	EXTRACT_EATER_NY	Downloads articles from Eater NY Atom feed	— (RSS feed)	BRONZE/WEBSCRAPING/
02	CLEAN_EATER_NY	NLP: clean, dedup, stopwords → summary_clean	bronze/webscraping/	SILVER/WEBSCRAPING/
03	EXTRACT_OPENTABLE	Playwright session → paginates reviews per restaurant	— (web)	BRONZE/OPENTABLE/
04	CLEAN_OPENTABLE	Flattens nested JSON → NLP → dedup by review_id	bronze/opentable/	SILVER/OPENTABLE/
05	ANALYZE_TRENDS	TF-IDF on articles+reviews → top 10 keywords → XCom	silver/ (ws + ot)	LIST[STR] KEYWORDS
06	SEARCH_SPOONACULAR	Queries Spoonacular with each XCom keyword	keywords (XCom)	BRONZE/API/
07	CLEAN_RECIPES	Flattens lists, strips HTML, casts types, dedup ID	bronze/api/	SILVER/API/
08	PREPARE_GOLD	Spark: transforms + governance KPIs + storytelling	silver/ (all layers)	GOLD/ (4 PARQUETS)



empire-tastes.nyc/airflow:8080



DAGs

Cluster Activity

Datasets

Security

Browse

Admin

Docs



15:24 -05 (-05:00)

AF

DAGs

All 4

Active 1

Paused 3

Running 1

Failed 0

Filter DAGs by tag

Search DAGs

Auto-refresh



DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Action
☐ bronze_ingestion_pipeline bronze	data_engineering_team		@daily		2026-06-03, 19:00:00		▶ 📄
☐ gold_processing_pipeline gold spark	data_engineering_team		@daily		2026-06-03, 19:00:00		▶ 📄
☒ nyc_gastronomy_pipeline bronze gold pipeline sequential silver	data_engineering_team	3	@daily	2026-06-05, 12:19:05	2026-06-04, 19:00:00	11	▶ 📄
☐ silver_processing_pipeline silver	data_engineering_team		@daily		2026-06-03, 19:00:00		▶ 📄

SILVER LAYER — CLEANING & NLP

TEXT PIPELINE — ARTICLES

1 REMOVE PHOTO CREDITS

2 DECODE HTML ENTITIES

3 STRIP URLS

4 LOWERCASE + NO PUNCTUATION

5 FILTER STOPWORDS

6 REMOVE SHORT/NUMERIC TOKENS

TRENDING KEYWORD EXTRACTION

CORE CHALLENGE

Food tokens vs. noise — how do we tell them apart?

HYBRID SOLUTION

- TF-IDF (sklearn) — distinctive tokens per batch
- spaCy vectors — cosine similarity vs. gastronomic anchor
- CUISINE_VOCAB — curated list for terms vectors miss

REVIEW STOPWORDS (OPENTABLE)

Generic opinion words filtered before Spoonacular query:

amazing · wonderful · beautiful · favorite · staff · service · dinner · experience · recommend...

GOLD LAYER — ANALYSIS & INSIGHTS

VADER SENTIMENT ANALYSIS

Compound score -1 to +1 · optimized for media & social text

$\geq +0.05 \rightarrow$ **POSITIVE**

-0.05 TO $+0.05 \rightarrow$ **NEUTRAL**

$\leq -0.05 \rightarrow$ **NEGATIVE**

KEYWORDS — TREND LABELS

TF-IDF SCORE	SENTIMENT	LABEL
≥ 80	POSITIVE	HIGHLY TRENDING
≥ 50	POSITIVE	TRENDING
≥ 30	POSITIVE	GROWING
≥ 50	NEGATIVE	HEAVILY CRITICIZED

10 AGGREGATIONS GENERATED IN GOLD

1	SENTIMENT_DISTRIBUTION	% positive/neutral/negative
2	SENTIMENT_TREND	Weekly mood evolution
3	TOP_KEYWORDS	Top 20 food keywords (TF-IDF)
4	KEYWORD_SENTIMENT	Sentiment per keyword
5	RECIPE_TREND_ALIGNMENT	Recipes aligned with trends
6	REVIEWS_SENTIMENT	Real diner sentiment
7	REVIEWS_BY_RESTAURANT	Satisfaction ranking
8	NAMED_ENTITIES	Restaurants, chefs, neighborhoods (NER)
9	DIETARY_BREAKDOWN	Recipes by dietary restriction
10	SOURCE_COMPARISON	Press vs recipes vs reviews

DEPLOYMENT — DOCKER

SERVICES IN DOCKER-COMPOSE

SERVICE	PORT	TECH	ROLE
AIRFLOW-SCHEDULER	—	AIRFLOW 2.10	EXECUTES DAGS
AIRFLOW-WEBSERVER	8080	AIRFLOW 2.10	MONITORING UI
GASTRONOMY-API	8000	FASTAPI	SERVES GOLD DATA
WEB	3000	NEXT.JS 15	USER DASHBOARD
GOVERNANCE-DASHBOARD	8050	PLOTLY DASH	TECHNICAL KPIS
STORYTELLING-DASHBOARD	8051	PLOTLY DASH	TRENDS VIEW
POSTGRES	5432	POSTGRESQL	AIRFLOW METADATA

VOLUMES & CONFIGURATION

DATALAKE_BRONZE/

Mounted as volume → DAGs write here

DATALAKE_SILVER/

Shared between Airflow and the API

DATALAKE_GOLD/

Shared between Airflow and FastAPI (read by API)

AIRFLOW/DAGS/

Hot-reload: DAGs update without rebuild

API — SERVICE LAYER

FASTAPI — LOCALHOST:8000

ENDPOINT	DESCRIPTION
<code>GET /API/STORYTELLING/SENTIMENT</code>	Combined article + review distribution
<code>GET /API/STORYTELLING/KEYWORDS</code>	Top keywords with score and sentiment
<code>GET /API/STORYTELLING/TREND</code>	Weekly mood evolution
<code>GET /API/STORYTELLING/ENTITIES</code>	Restaurants, chefs, neighborhoods (NER)
<code>GET /API/STORYTELLING/SOURCES</code>	Articles with links + reviews per restaurant
<code>GET /API/STORYTELLING/SUMMARY</code>	Top keyword, % positive, total recipes
<code>GET /API/RECOMMENDATIONS/</code>	Recipes filtered by cuisine/diet/time
<code>GET /API/GOVERNANCE/KPIS</code>	Data quality KPIs
<code>GET /API/GOVERNANCE/NULL-RATES</code>	Completeness per field

IMPLEMENTATION NOTES

LOADER DESIGN

api/loader.py reads Gold Parquet directly — auto-detects local (Docker) vs. S3 (AWS Lambda).

AUTOMATIC PARTITIONING

Always serves latest partition. Pipeline down today?
Yesterday's data — no failure.

SENTIMENT FIX

/sentiment merges Eater NY + OpenTable Gold sources —
blending press vs. diner opinion.



empire-tastes.nyc/api:8000/docs#/

Empire's Taste API

1.0.0 OAS 3.1

/openapi.json

NYC Gastronomic Trends & Sentiment — Gold layer REST interface

Governance



GET

/api/governance/ Get Governance



GET

/api/governance/kpis Get Kpi Summary



GET

/api/governance/null-rates Get Null Rates



Storytelling



GET

/api/storytelling/ Get Storytelling



GET

/api/storytelling/sentiment Get Sentiment



GET

/api/storytelling/keywords Get Keywords



GET

/api/storytelling/trend Get Sentiment Trend



GET

/api/storytelling/entities Get Named Entities



GET

/api/storytelling/summary Get Narrative Summary



GET

/api/storytelling/sources Get Sources



DATA GOVERNANCE

The spark_governance.py script measures 8 quality dimensions on Silver data and persists them in Gold in long format (one row per KPI).

COMPLETENESS

% of null fields per column.
Healthy threshold: < 20%

VOLUME

Total records and Parquet
files generated per day

UNIQUENESS

% duplicates by primary key
(recipe ID or article_id)

VALIDITY

% expected columns
present + IQR outliers

ACCURACY

NLP compression ratio + %
non-ASCII chars (noise proxy)

TIMELINESS

Number of daily partitions
present in Bronze

OUTPUT FORMAT — LONG

One Parquet: kpi_name · category · source
field · value · unit · finding
Queryable with any tool — no extra transforms.

Exposed in FastAPI: GET /api/governance/kpis · GET /api/governance/null-rates · GET /api/governance/outliers



Empire's Taste — NYC Food Trends

Powered by Eater NY & Spoonacular | Last updated: 2026-06-05 17:21:30

This Week in NYC Food

This week, "chicken" is the most discussed food topic in NYC media, with 100% positive coverage. 45 matching recipes are available in the database. Gluten-free: 60.0% of recipes are trending.

NYC Mentions — Who & Where

Restaurants

- The Best New Restaurants (2)
- Noortwyck (1)
- Wood-Fired Sibling (1)
- the American Dream Mall (1)
- Standout Omakases Celebrates (1)

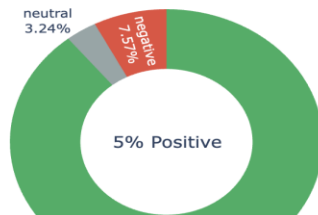
Neighborhoods

- New Jersey (1)
- NYC (1)
- Jersey City (1)
- Queens (1)
- Brooklyn (1)

Chefs

—

How NYC Food Media Feels This Week



- positive
- negative
- neutral

Opinion Shifts Over Time

Avg Compound Score

1
0.5
0
-0.5





empire-tastes.nyc/dashboard:8050

Empire's Taste — Data Governance Dashboard

Team: Data Engineering | Last updated: 2026-06-05 17:21:08

Total Records

100

Max Null Rate

100.0%

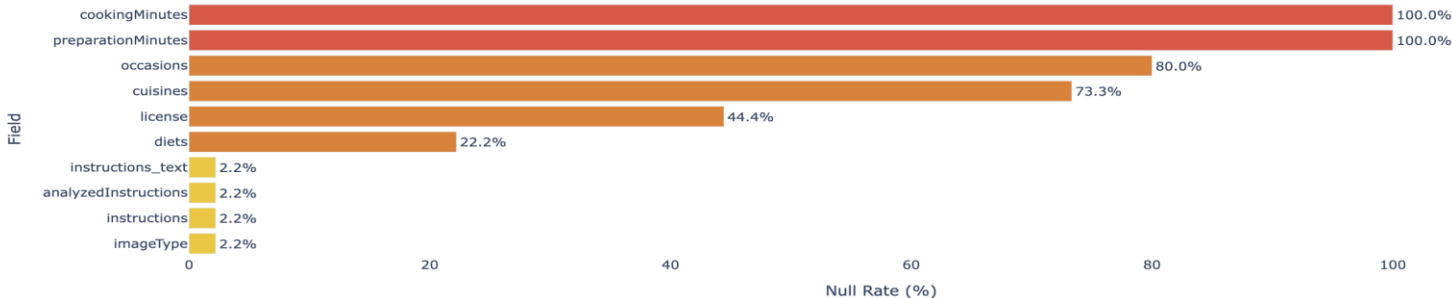
Duplicate Rate

50.00%

Schema Compliance

100.0%

Null Rate per Field — Silver Layer



Records Ingested per Day by Source



Outlier Rate per Numeric Field (IQR Method)





WHAT LIES BEYOND THE DASHBOARD?



We recognize that data alone does not create value. The business requires a strategic framework that translates technical information into real-world utility.



empire-tastes.nyc/home

GASTRONOMIC INTELLIGENCE

NEW YORK CITY

● updated daily

EMPIRE'S TASTE.

THE CITY IS EATING SOBA

Eater NY articles + Spoonacular recipes — processed
through a sentiment pipeline daily. Read the street like
data.

ENTER DASHBOARD →

— 5,000 recipes · updated daily —

THE WEB APP



empire-tastes.nyc/home



EMPIRE'S
TASTE

WELCOME
BACK.

One step. Pick your Google account and walk in.

 CONTINUE WITH GOOGLE

no passwords · no forms

THE WEB APP

/DASHBOARD/TRENDS

Hero with NYC video + dark background

Keyword of the day: "The city is eating bakery" — connected to live API

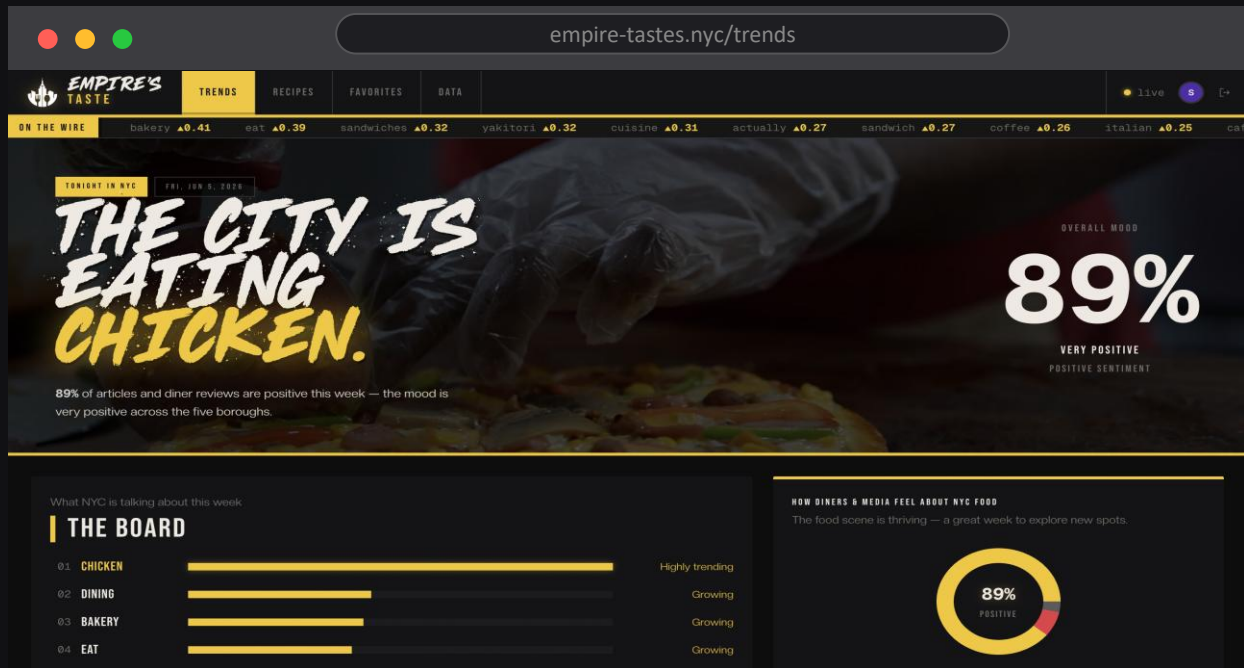
Positive sentiment % (89%) + mood label

THE BOARD: keywords with human labels (Highly Trending, Growing)

Sentiment donut: articles + reviews combined

Weekly trend line (grows with daily data)

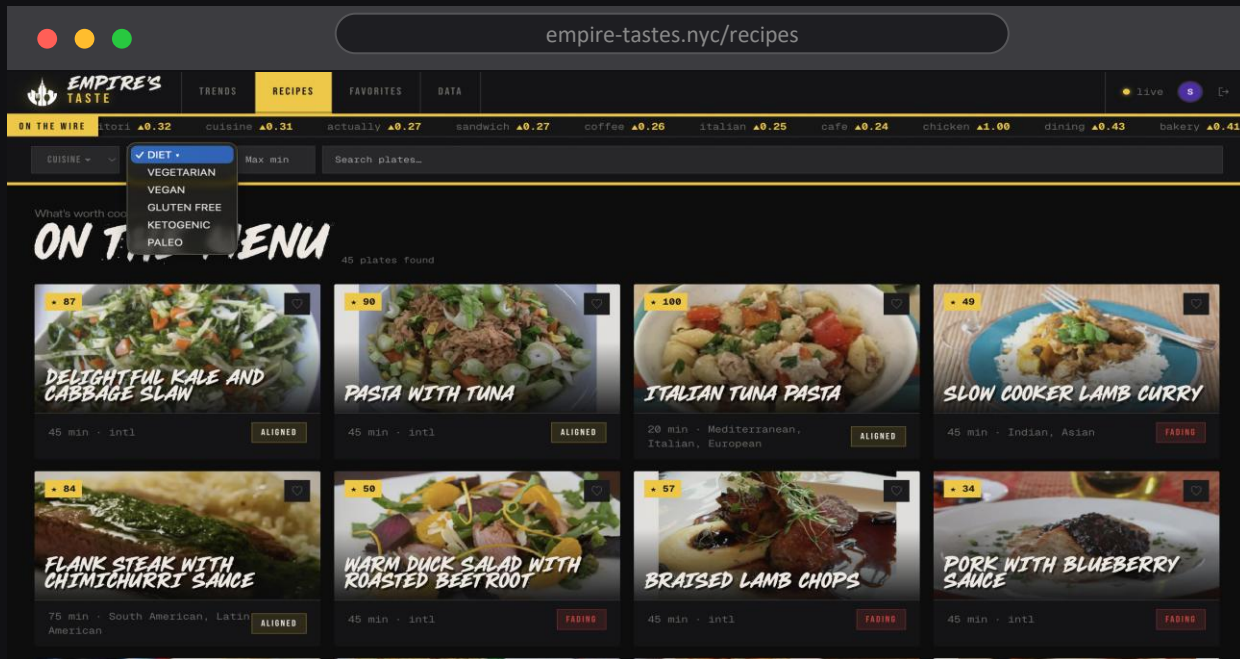
Entity table: restaurants, chefs, neighborhoods detected by NER



THE WEB APP

/DASHBOARD/RECIPES

- Trending recipes from Spoonacular selected by today's keywords
- Filters: cuisine, diet, prep time
- Cards with image, health score, time, ingredients
- Favorites system saved per user (Prisma + SQLite)
- /dashboard/favorites — personal collection of saved recipes
- Persists per authenticated user (Google OAuth)





empire-tastes.nyc/recipes/641422



EMPIRE'S
TASTE

TRENDS

RECIPES

FAVORITES

DATA

live

S

[→]

ON THE WIRE

italian ▲0.25

cafe ▲0.24

chicken ▲1.00

dining ▲0.43

bakery ▲0.41

eat ▲0.39

sandwiches ▲0.32

yakitori ▲0.32

cuisine ▲0.31

actually ▲0.2

DELIGHTFUL KALE AND CABBAGE SLAW

Ready in

45 min

Servings

6

Health score

87/100

Likes

2

GLUTEN FREE

DAIRY FREE

INGREDIENTS

- cabbage
- kale
- celery
- scallion
- carrot
- dressing
- red wine vinegar
- mustard
- agave nectar
- salt
- olive oil

INSTRUCTIONS

In a very large bowl, combine cabbage, kale, celery, scallion and carrot. Toss well. In a small bowl, combine dressing ingredients. Pour dressing over vegetables and toss to coat.



empire-tastes.nyc/favorites



TRENDS

RECIPES

FAVORITES

DATA

live

S

[+]

ON THE WIRE

sandwiches ▲0.32

yakitori ▲0.32

cuisine ▲0.31

actually ▲0.27

sandwich ▲0.27

coffee ▲0.26

italian ▲0.25

cafe ▲0.24

chicken ▲1.00

dining .

Your collection

SAVED PLATES.

2

recipes saved

DOWNLOAD PDF ↓

Saved recipes expire after 30 days. Download any you want to keep.



ITALIAN TUNA PASTA

Expires in 30d

REMOVE ×



DELIGHTFUL KALE AND
CABBAGE SLAW

Expires in 30d

REMOVE ×



empire-tastes.nyc/download

Saved Plates

2 recipes · Empire's Taste NYC · June



MEDITERRANEAN, ITALIAN, EUROPEAN

Italian Tuna Pasta

READY IN
20 min

DAIRY FREE

INGREDIENTS

- chillies
- juice of lemon
- bell pepper

INSTRUCTIONS

Once pasta is cooked, drain and leave to cool for a minute. Place small skillet on medium fire, drizzle olive oil, add in red pepper and stir-fry for 1-2 minutes. Put aside. Toss pasta shells, red pepper, tuna, parsley, garlic, chillies and lemon juice. Season with ground black pepper to taste, spoon into serving bowls.

about blank

1/3

5/26, 11:25

Empire's Taste - Saved Plates



Italian Tuna Pasta

READY IN
20 min

SERVINGS
3

HEALTHY
100/100

LIKES
1

DAIRY FREE

INGREDIENTS

- chillies
- garlic
- ground pepper
- juice of lemon
- parsley leaves
- pasta shells
- bell pepper
- tuna

INSTRUCTIONS

Once pasta is cooked, drain and leave to cool for a minute. Place small skillet on medium fire, drizzle olive oil, add in

- parsley leaves
- tuna

- pasta shells

Imprimir

3 páginas

Destino

Guardar como PDF

Páginas

Todas

Páginas por hoja

1

Márgenes

Predeterminados

Opciones

- ☒ Encabezado y pie de página
- ☐ Gráficos de fondo

Imprimir utilizando el cuadro de diálogo del sistema (⌘P)



Vista previa de PDF



Cancelar

Guardar

THE WEB APP

/DASHBOARD/DATA

KPIs in plain English: "Records collected", "Missing data", "Data quality"

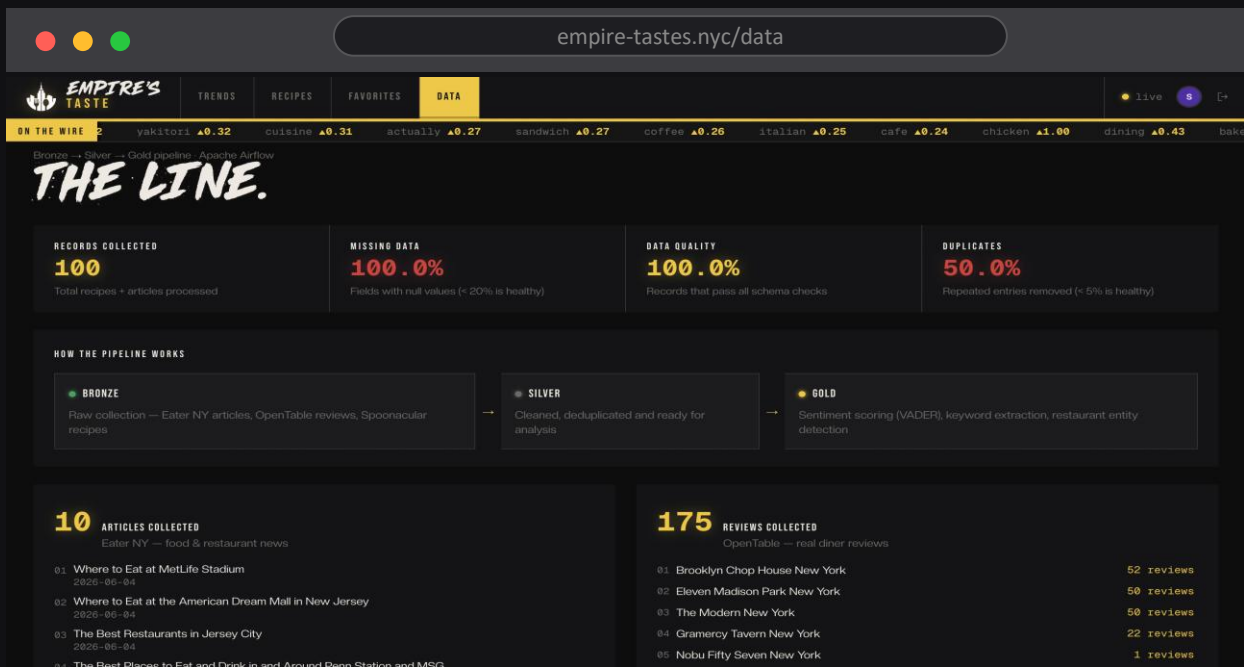
Pipeline layers explained in simple language

Article counter with clickable titles → Eater NY

Reviews per restaurant (OpenTable)

Data completeness grid — field + Healthy/Review needed badge

Authentication: Google OAuth (NextAuth.js) — no passwords





empire-tastes.nyc/data



TRENDS

RECIPES

FAVORITES

DATA

live

S



ON THE WIRE

27

coffee ▲0.26

italian ▲0.25

cafe ▲0.24

chicken ▲1.00

dining ▲0.43

bakery ▲0.41

eat ▲0.39

sandwiches ▲0.32

yakitori ▲0.32

cuisine

DATA COMPLETENESS BY FIELD

Shows what percentage of data is missing for each field. Under 20% is healthy — the higher the bar, the more records lack that information.

Recipe prep time



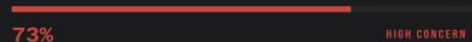
Recipe cooking time



Occasion tags (e.g. holiday, party)



Cuisine type (e.g. Italian, Japanese)



Recipe license info



Dietary labels (vegan, keto...)

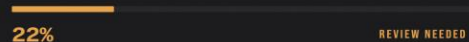
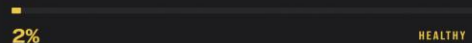
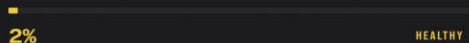


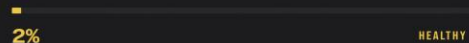
image type



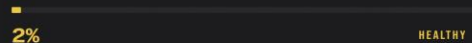
instructions



analyzed instructions

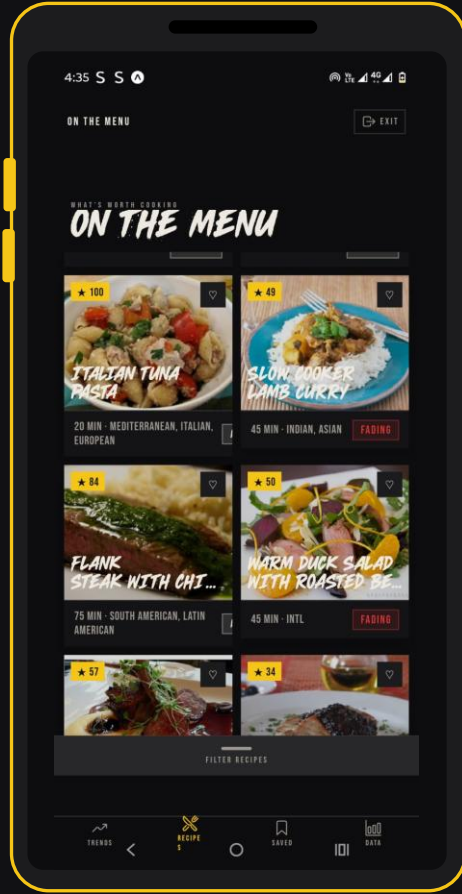


instructions_text





THE MOBILE APP

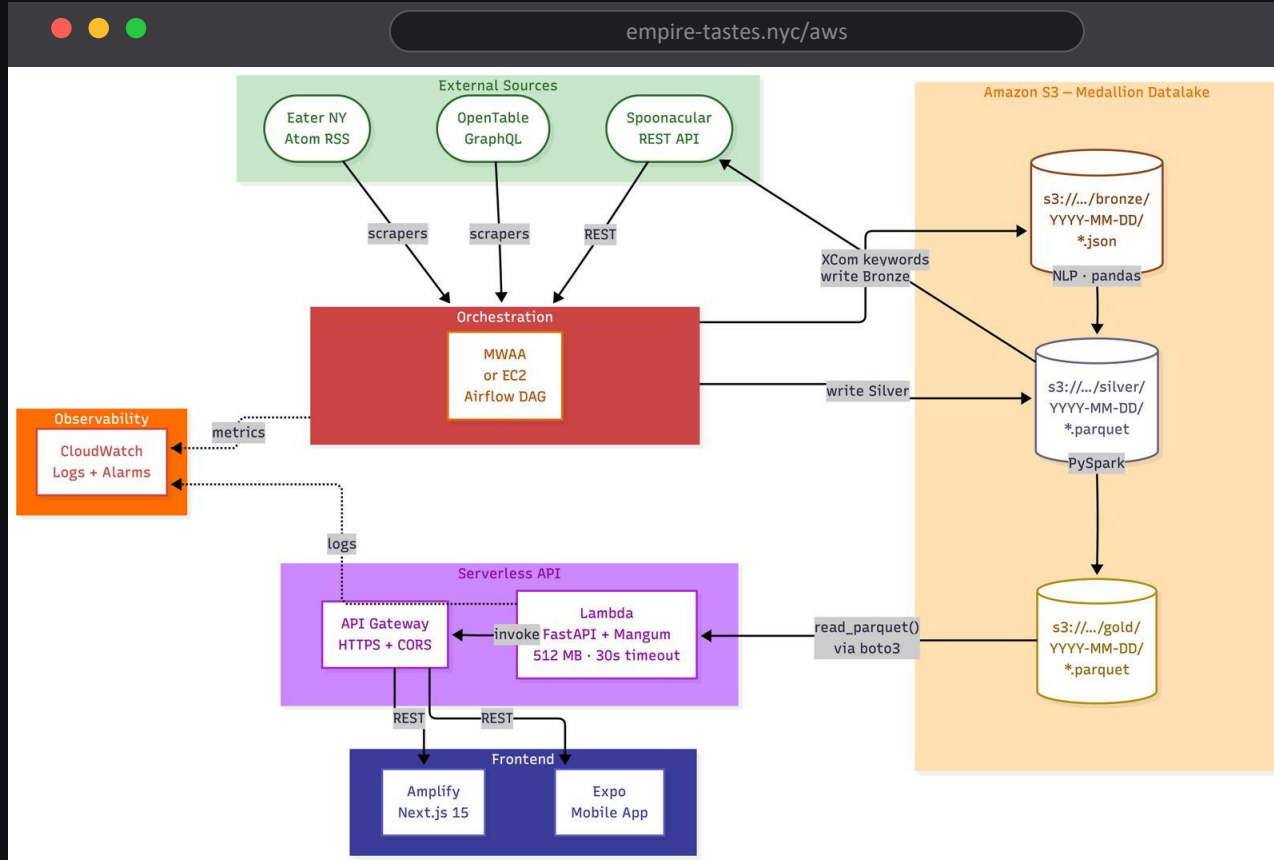




SCALING TO PRODUCTION: SCALABILITY AND RESILIENCE ON AWS



Designing a robust infrastructure based on the AWS Well-Architected Framework to ensure business continuity.



THANKS, YOU

**THAT WAS
EMPIRE'S TASTE**